SIMATS ENGINEERING ASSESSMENT

TOOL 1

COURSE NAME: INTERNET PROGRAMMING **Shanmukha.B**

COURSE CODE: CSA4303 **192372217**

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly. **Questions:**

1. Identify appropriate **HTML5 semantic elements** missing in the structure.

Ans : Semantic elements such as <header>, <main>, and <section> are missing. They provide meaningful structure and improve accessibility.

2. Modify the structure to make it **standards-compliant**.

Ans : Use a valid HTML5 document structure with <!DOCTYPE html>, proper metadata, labels, and semantic elements. Close all tags correctly.

3. Interpret the **HTTP request generated** during form submission.

Ans : The POST method sends the form data to the /register server endpoint. The data is included in the request body rather than the URL.

4. Analyze the **HTTP response cycle** from server to browser.

Ans : The server receives and processes the request, then sends an HTTP response with a status code and content. The browser displays the resulting page or message.

5. Suggest improvements for **usability and accessibility**.

Ans : Add labels, required-field validation, clear instructions, and meaningful error messages. Use keyboard-accessible controls and appropriate input types.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body> </html>
```

Questions:

1. Identify improper nesting and missing closing tags.

The <h1> and <p> elements are missing their closing tags. The original HTML also has an incomplete structure that should be corrected.

2. Analyze how different browsers may interpret this structure.

Browsers may automatically correct malformed HTML differently, potentially causing inconsistent layouts. This can affect the display and document structure.

3. Identify missing semantic elements (e.g., header, section).

The original code lacks semantic elements such as <header>, <main>, and <section>. These elements improve page organization and accessibility.

4. Provide a corrected W3C-compliant HTML structure.

Use <!DOCTYPE html>, proper nesting, and matching opening and closing tags. Include valid HTML5 semantic elements and required metadata.

5. Suggest techniques to ensure cross-browser compatibility.

Use valid HTML and CSS, test the website in different browsers, and use CSS resets when needed. Follow web standards and use responsive design techniques.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the form configuration.

The button uses type="button" instead of type="submit", so it does not submit the form automatically. The GET method is also unsuitable for transmitting passwords.

2. Analyze why the HTTP request is not triggered.

The type="button" control does not trigger form submission by default. Changing it to type="submit" allows the browser to submit the form.

3. Interpret the role of GET method in this scenario.

The GET method sends form data through URL query parameters. It is unsuitable for passwords because sensitive information may appear in browser history, URLs, and logs.

4. Propose a corrected implementation.

Use method="POST" and type="submit" in the form. Include named input fields and appropriate validation to submit the data correctly.

5. Suggest improvements for secure data transmission.

Use HTTPS to encrypt data during transmission and validate input on the server. Store passwords securely using a strong password-hashing algorithm.

Code of Conduct:

I _____Shanmukha.B(192372217)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shanmukha.B

RUBRICS FOR CALCULATION:

Criteria	Weig htage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandin g of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptio n s	Poor understandin g; major misconcepti o ns
Explanation	25%	Detailed, wellexplained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1.Online Symposium Portal

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Symposium Registration</title>
</head>
<body>
    <header>
        <h1>Online Symposium Portal</h1>
    </header>

    <main>
<section>
    <h2>Student Registration</h2>
    <form action="/register" method="POST">
        <label for="name">Name:</label>
        <input type="text" id="name" name="name" required>

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>

        <button type="submit">Submit</button>
    </form>
</section>
    </main>
</body>
</html>
```

Output :

Online Symposium Portal

Student Registration

Name: Email: Submit

2. Cross-Browser Compatibility Issue

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Company Website</title>
</head>
<body>
    <header>
        <h1>Welcome</h1>
    </header>

    <main>
        <section>
            <h2>About Us</h2>
            <p>Welcome to our company website.</p>
        </section>
    </main>
</body>
</html>
```

Output :

Welcome

About Us

Welcome to our company website.

3. Form Submission Failure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login Form</title>
</head>
<body>
    <header>
        <h1>User Login</h1>
    </header>

    <main>
        <section>
            <form action="/submit" method="POST">
                <label for="username">Username:</label>
                <input type="text" id="username"
```

```
            name="username" required>

        <label for="password">Password:</label>
        <input type="password" id="password"
            name="password" required>

        <button type="submit">Login</button>
    </form>
        </section>
    </main>
</body>
</html>
```

Output :

